One-To-One Functions

A function $f: A \to B$ associates a single $b \in B$ to each $a \in A$, but the same $b \in B$ may be associated to several values of A.

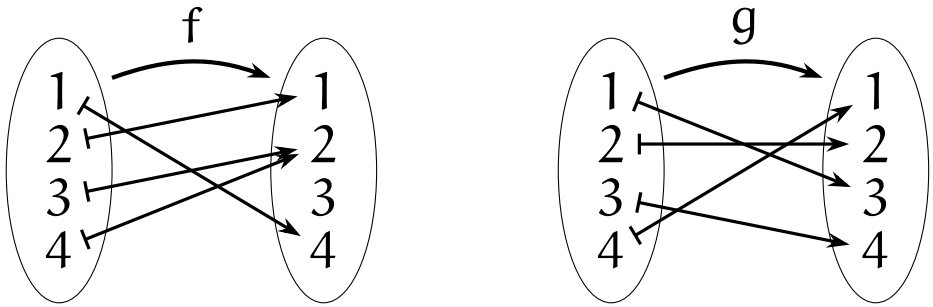
For example, for $f(x) = x^3 - x$ we have $f(-1)=0$ and $f(0)=0$ and $f(1) = 0$.

If f does *not* assign the same value to different inputs, it is called ***one-to-one*** (1-1), or ***injective***:

$$\forall\, a, a' \in A \left(a \neq a' \;\Rightarrow\; f(a) \neq f(a') \right).$$

`For all a ∈ A, a' ∈ A, if a ≠ a' then f(a) ≠ f(a').`

Example: Consider the following functions:



f is not injective, as $f(3) = f(4) = 2$.

However, g is injective.

One-To-One Functions

Exercise: Which of the following functions are one-to-one?

1. The function $\text{score}: \text{Class} \to \text{Marks}$. Not one-to-one because there can be repeats.

2. The function $f: \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$. One-to-one because it's not possible to have repeats.

3. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$. Not one-to-one because there can be repeats. 2^2 and (-2)^2 both produce 4.

4. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$. One-to-one, because you can't have repeats.

x^2 and x^3 are different because two minus symbols cancel each other out:
-2 * -2 = 4
whereas,
-2 * -2 * -2 = -8

One-to-one was about repeats. Onto is about gaps.

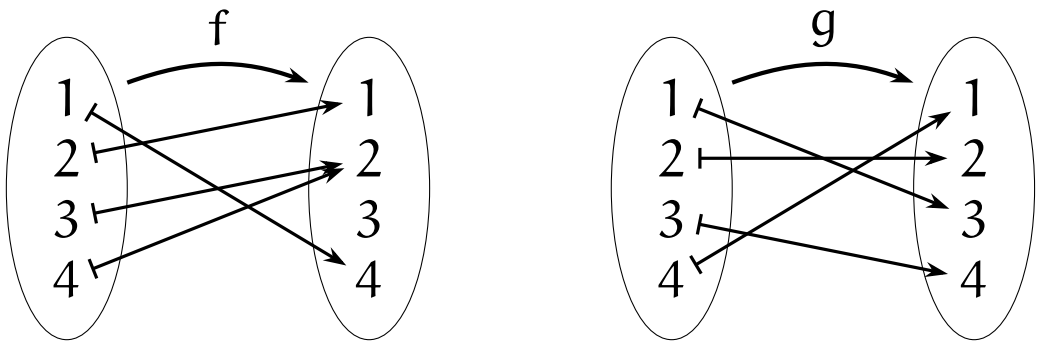
Onto Functions

A function $f: A \to B$ is ***onto***, or ***surjective***, if, and only if, every value in its codomain is produced by the function; that is, its range equals its codomain:

$$\forall\, b \in B\ \exists\, a \in A\ \Big(f(a) = b\Big).$$

`For all b ∈ B there exists a ∈ A such that f(a) = b.`

Example: Consider the following functions:



f is not surjective, as the value 3 is not produced by the function.

However, g is surjective.

Onto Functions

Exercise: Which of the following functions are onto?

1. The function $\text{score}: \text{Class} \to \text{Marks}$.

Not onto
Student 1 -> 65
Student 2 -> 72
100 <- nobody got this.

2. The function $f: \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$.

Not onto.
1^1 = 1
2^2 = 4
... what about 2, 3? never got hit.

3. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

Not onto.
Negative numbers never get hit.
Squaring negative numbers always give a positive

4. The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$.

Onto
Every number can be hit because cubing can result in both negative and positive numbers – real numbers include decimals.

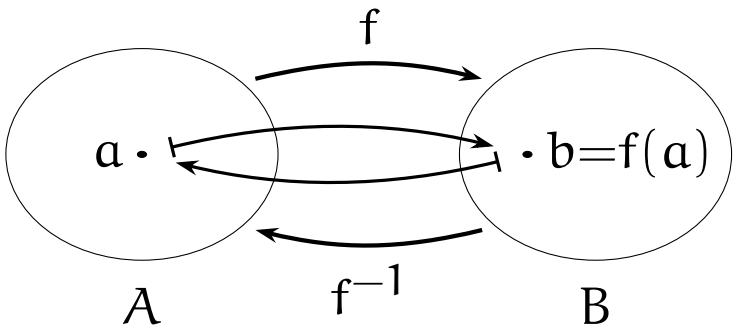
Although 2^3 and 3^3 skip numbers such as 2, 3,
you can use 1.26^3 to hit 2 and 1.44^3 to hit 3.

Bijective Functions

A function $f\colon A \to B$ is ***bijective***, or a ***bijection***, if, and only if, it is both one-to-one and onto.

The arrows of a bijection can be reversed to give the ***inverse function*** $f^{-1}\colon B \to A$ that assigns to each $b \in B$ the unique $a \in A$ such that $f(a)=b$:

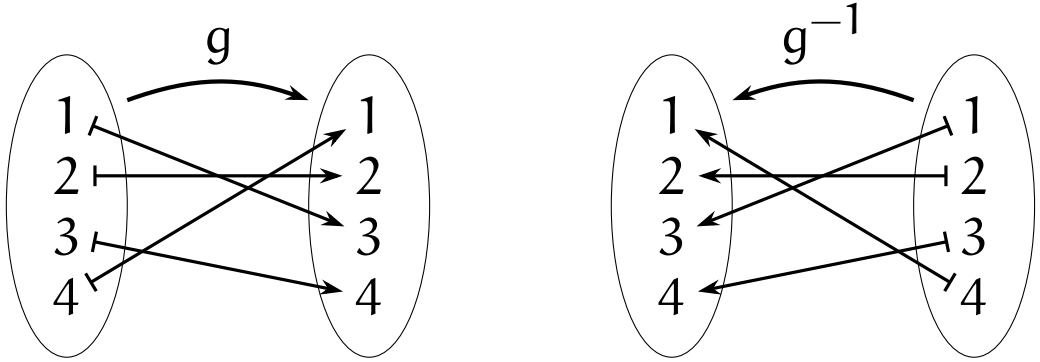
$$f^{-1}(b) = a \text{ if, and only if, } f(a)=b$$



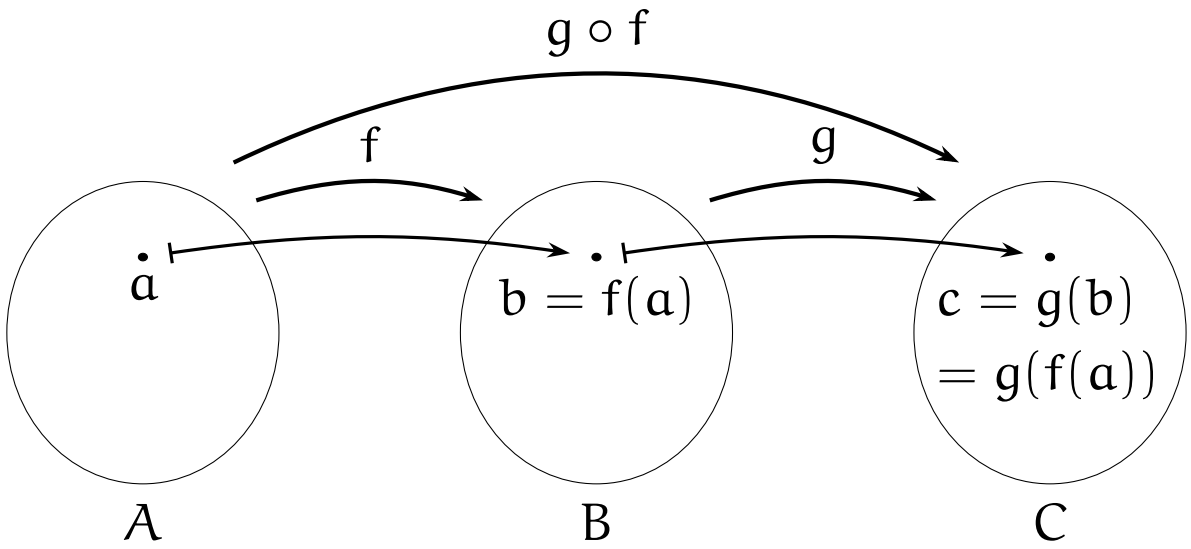
$f^{-1}\colon B \to A$ is also a bijection, and $(f^{-1})^{-1} = f$.

Example:

Every output is hit, and there are no repeats.
This is bijective as it's one-to-one, and onto.



Properties of Composed Functions



- If $f : A \to B$ and $g : B \to C$ are both one-to-one,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both onto,

  then so is $g \circ f : A \to C$.

- If $f : A \to B$ and $g : B \to C$ are both bijective,

  then so is $g \circ f : A \to C$.